ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-1 **WINTER SEMESTER, 2024-2025**

DURATION: 40 MINUTES **FULL MARKS: 20**

CSE 4309: Theory of Computing

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. What is the formal definition of a Finite Automaton? 3 (CO1) (PO1)

Solution:

A finite automaton D is 5-tuple:

$$D = \langle Q, \Sigma, \delta, q_0, F \rangle$$

- Q is a finite set of **states**
- Σ is a finite set of **symbols**
- $\delta : Q \times \Sigma \to Q$ is the **transition function**
- $q_0 \in Q$ is the **initial state**
- $F \subseteq Q$ is the set of **final states**

Rubric:

- 0.5 point for each correct aspect of the definition.

2. Prove that the class of regular languages is closed under the union operation. 4 (CO2) (PO1)

Solution:

We have to prove that, if L_1 and L_2 are regular languages, then $L_1 \cup L_2$ is also a regular language. Let the finite automata M_1 and M_2 accept L_1 and L_2 respectively.

$$M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

$$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

where $L(M_1) = L_1$ and $L(M_2) = L_2$.

We will construct a new FA, $M = (Q, \Sigma, \delta, q_0, F)$, that accepts $L_1 \cup L_2$. This machine simulates the operations of M_1 and M_2 in parallel.

The components of M are defined as follows:

1. **States (Q):** The set of states is the Cartesian product of the sets of states of M_1 and M_2.

$$Q = Q_1 \times Q_2 = \{(r, s) \mid r \in Q_1, s \in Q_2\}$$

2. **Alphabet (Σ):** The input alphabet remains the same.

3. **Transition Function (δ):** For any state $(r, s) \in Q$ and input symbol $a \in \Sigma$, the transition is defined by applying the original transition functions to each component:
$$\delta((r, s), a) = (\delta_1(r, a), \delta_2(s, a))$$

4. **Start State (q_0):** The start state is the pair consisting of the start states of M_1 and M_2.
$$q_0 = (q_1, q_2)$$

5. **Final States (F):** Since we want to accept the union ($L_1 \cup L_2$), a state in M is accepting if *at least one* of the component states is an accepting state in its original machine.
$$F = \{(r, s) \in Q \mid r \in F_1 \vee s \in F_2\}$$
Alternatively, this can be expressed using set operations as:
$$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$$

For any input string $w = w_1 w_2 \ldots w_n$, let the computation of M_1 on w end in state r_n and the computation of M_2 on w end in state s_n. By our definition of δ, the computation of M on w will end in state (r_n, s_n).

M accepts w if and only if $(r_n, s_n) \in F$. Based on our definition of F:
$$(r_n, s_n) \in F \iff r_n \in F_1 \text{ or } s_n \in F_2$$
This is equivalent to saying:
$$w \in L(M_1) \text{ or } w \in L(M_2)$$
$$w \in L_1 \cup L_2$$

Since we have constructed a finite automaton M that accepts $L_1 \cup L_2$, the language $L_1 \cup L_2$ is regular. $\square$

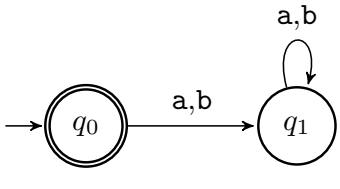
Rubric:

- 4 points for a correct proof.

3. Create finite automata for the following languages: $1 + 3 + 4$ (CO3) (PO1)

a) $L =$ the language that accepts only the empty string ϵ with $\Sigma = \{\mathbf{a}, \mathbf{b}\}$

Solution:

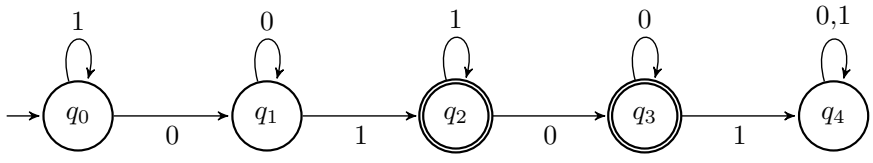


Rubric:

- 1 point for a correct FA.

b) $L = \{w \in \{0,1\}^* \mid w$ contains exactly one occurrence of the substring $01\}$

Solution:

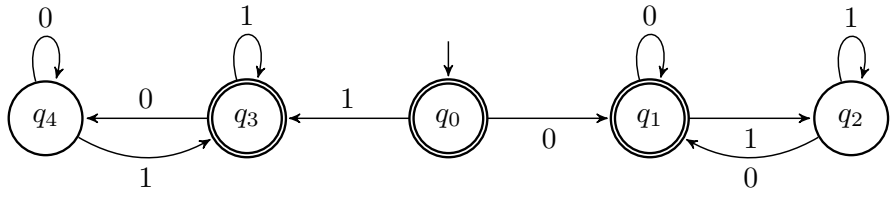


Rubric:

- 3 points for a correct FA.

c) $L = \{w \in \{0,1\}^* \mid w$ contains an equal number of occurrences of the substrings 01 and $10\}$

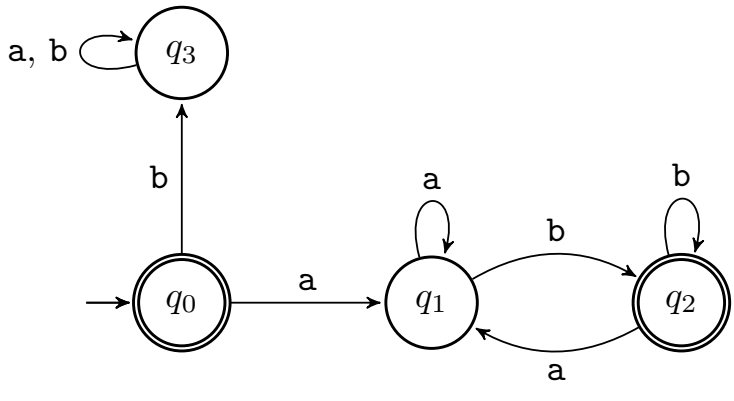
Solution:



Rubric:

- 4 points for a correct FA.

4. Consider the following finite automaton D with $\Sigma = \{\mathbf{a}, \mathbf{b}\}$: $2+3$ (CO3) (PO1)



a) Let the extended transition function $\delta^* : Q \times \Sigma^* \to Q$ of D be defined as:

- **(Basis Case)** $\delta^*(q, \epsilon) = q$
- **(Induction Case)** $\delta^*(q, cw) = \delta^*(\delta(q, c), w)$ where $c \in \Sigma$, $w \in \Sigma^*$

Determine the results of the following extended transition functions:

i. $\delta^*(q_1, \epsilon)$
ii. $\delta^*(q_0, \mathbf{bab})$
iii. $\delta^*(q_2, \mathbf{abab})$
iv. $\delta^*(q_1, \mathbf{aaabbba})$

Solution:

i. $\delta^*(q_1, \epsilon) = q_1$

ii. $\delta^*(q_0, \mathbf{bab}) = q_3$

iii. $\delta^*(q_2, \mathbf{abab}) = q_2$

iv. $\delta^*(q_1, \mathbf{aaabbba}) = q_1$

Rubric:

- 0.5 point for each correct destination state.

b) Determine the language $L(D)$ accepted by D and provide a brief explanation.

Solution:

$$L(D) = \{w \in \{\mathbf{a}, \mathbf{b}\}^* | w \text{ is an empty string } \epsilon \text{ or } w \text{ starts with } \mathbf{a} \text{ and ends with } \mathbf{b}\}$$

Rubric:

- 3 points for a correct language description.